Xinyao Rao

Autonomous Driving | Foundation Models | Learning & Planning

744044292@qq.com | github.com/Rtwotwo

EDUCATION

Nanjing University Sep 2026 - Sep 2029 (expected)

Graduate studies | *Autonomous Driving & Foundation Models*
CITE Lab | 3D Vision Lab (NJU-3DV)
Advisors: Xun Cao and Wei Li

Northwestern Polytechnical University Sep 2022 - Jun 2026

Undergraduate studies

SELECTED ACHIEVEMENT

NAVSIM Dual-Leaderboard #1 As of Nov 5, 2026

Ranked **1st as EABOT.AI**: NAVTEST v1 **96.1906 PDMS**; NAVHARD v2 **64.5806 EPDMS**.

RESEARCH & ENGINEERING PROJECTS

DriveTTO | Test-Time Trajectory Optimization Project / Results

- Developed **training-free trajectory optimization** for frozen driving planners, retaining a complete reference plan while exploring endpoint, cross-planner, and bounded control-residual proposals.
- Used **reference-relative selection** to compare candidates; built NAVSIM visualizations and a HUGSIM closed-loop rollout gallery to inspect search decisions and driving behavior.
- Reached **95.6055 PDMS on NAVTEST (#3)** and **58.6340 EPDMS on NAVHARD (#5)** with the official EABOT.AI&NJU entry linked to DriveTTO.

CoFiDrive | Camera-Based End-to-End Planning Code

- Extended a DrivoR-based planner with **DINOv2 / LoRA**, six scene-conditioned experts over 64 camera registers, and gated fusion of scene and spatial patch memories for trajectory scoring.
- Added **192 training-only control-space augmentation slots** to expand scorer supervision; kept inference at 64 proposals with no extra learned parameters from augmentation.
- Reported **92.4 PDMS** on NAVSIM-v1 navtest (train85k, 10 epochs, mean of three training seeds); released training, evaluation, ablation configurations, and planning visualizations.

DriveRSI-R0 | Controlled Policy Improvement Code

- Built **DriveHarness** to connect failure diagnosis, training-experience retrieval, branch generation, sample admission, native model training, and evaluation with explicit budgets and data provenance.
- Integrated model adapters, checkpoint recovery, and promotion / rollback gates with train/test separation and image-object-trajectory consistency checks; documented **191 selected CPU tests** in release verification.

DriveTraj | NAVSIM Trajectory Planning Leaderboard

- Submitted **DriveTraj-scale (95.5520 PDMS, #6)** and **DriveTraj-base (94.9290 PDMS, #10)** to the official NAVSIM-v1 NAVTEST leaderboard.

TECHNICAL SKILLS & OPEN SOURCE

Programming & Tools: Python, PyTorch, NumPy, OpenCV, Git, Linux
Modeling: Vision Transformers, DINOv2, LoRA / adapters, trajectory scoring, reward modeling, PPO / GAE
Research Engineering: NAVSIM evaluation, experiment recovery, data provenance, reproducible ablations

Open Source: Curate AutoDrive Research: source-linked papers, code, and evaluation protocols for end-to-end driving, VLA, and world models.